A Mechanized Metatheory for Mergeable Replicated Data Types:

RA-linearizability under three-way merge — eight verification conditions, a conditioned generalization, and the tombstone-free RGA end-to-end

Sal metatheory note
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Abstract

Mergeable replicated data types (MRDTs) resolve concurrent divergence by a *three-way* merge $\mathsf{mergeL}(l, a, b)$: the two divergent states a, b are reconciled relative to their lowest common ancestor l in a git-like version DAG. The Neem methodology reduces RA-linearizability of an MRDT to a fixed catalogue of per-data-type verification conditions (VCs), tied together by a once-and-for-all soundness meta-theorem. Mechanizing that meta-theorem in Lean 4, we find that its central induction is unsound as written — a reachable, fully LCA-legal execution of a legal add-wins MRDT defeats every bottom-up peel the proof can make — that the paper's own LCA lemma, while true, has a gap in its proof, and that no repair can lean on either of the two obvious crutches: the LCA argument cannot be forgotten (the counter MRDT provably admits no binary, CRDT-style collapse), and no lattice-flavoured contract is available (the lattice laws are equations over free tuples — the wrong instances for an LCA-aware merge — and no usable order survives). What works is a *delta* discipline: three unconditional relational VCs on `rc`, one merge symmetry, three *feasibility-conditioned* redistribution laws, and a single contextual *causal-delta* equation — eight VCs in all — from which every reachable configuration of the replicated store is RA-linearizable, *per version*. All of it is mechanized in Lean 4 with zero `sorry`s, and the VC set is validated in the strongest currency available: eleven of the twelve production MRDTs of the Sal repository — OR-sets, an enable-wins flag, counters, grow-only set and map, a tombstone RGA, Peritext, the multi-valued register and the add-wins priority queue — carry kernel-checked end-to-end RA-linearizability theorems through it. The one data type provably *outside* the VC set — the tombstone-free RGA, whose commutation holds only on reachable states — is hosted by a *conditioned* extension of the metatheory grown from this note's own open questions: a kernel-checked end-to-end theorem gives RA-linearizability *up to observational equivalence* at every reachable configuration, from a single honest-delivery assumption. This note is the complete, self-contained account: what an MRDT is and how to describe one (§2), the corrected flat metatheory and its validation (§3–§9), the conditioned metatheorem with its pen-and-paper end-to-end proof for the tombstone-free RGA (§10–§14), the open questions, and the mechanization map.

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1 Introduction

An MRDT execution looks like a git history: replicas fork, apply operations locally, and merge. Unlike a state-based CRDT, whose binary merge must reconcile two states with no memory of their common past, an MRDT's merge receives a third argument — the state at the *lowest common ancestor* (LCA) of the two versions being merged — and may use it to compute *deltas*: what each side did since the fork. The paradigmatic example is the counter,

$$\mathsf{mergeL}(l, a, b) \ = \ a + b - l,$$

which adds the two sides' increments-since-l instead of double-counting their common past (Figure 1).

The Neem methodology (OOPSLA 2025) promises a clean division of labour: the data-type designer discharges a fixed catalogue of equations over do, mergeL and the conflict-resolution relation rc — mostly by SMT — and a soundness meta-theorem, proved once, converts those VCs into *RA-linearizability* of every reachable configuration. This note documents what a Lean 4 mechanization of that meta-theorem, in the full **ternary, version-DAG** setting, found: the meta-theorem's central induction is unsound as written, and for LCA-sensitive data types several of the catalogue's merge VCs are false in principle when quantified over all states. It also develops

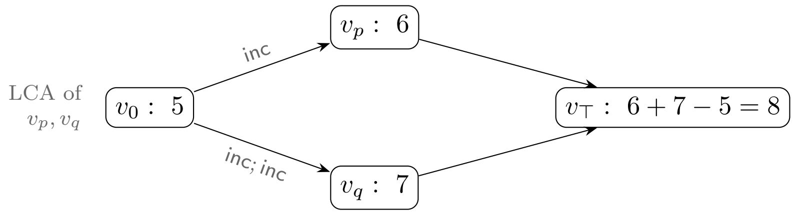


Figure 1: The counter MRDT. Both branches inherit the 5 from the fork point; the LCA argument lets the merge count each branch's *delta* exactly once. No binary merge of 6 and 7 alone can recover 8: the information "how much is shared" lives in the DAG, and the LCA argument is how the data type reads it.

the corrected metatheory: a set-relative linearization order, canonical states $\sigma(E)$ (each event set determines a unique state), and a ternary *Join Lemma* whose induction consumes a small, contextual VC surface. Its contributions:

1. **The paper's LCA lemma is true but its proof has a gap** (§4.1): $L(v_\ell) = L(v_1) \cap L(v_2)$ holds, but only as a *reachability invariant* of the store, whose proof needs a generator-uniqueness induction the paper silently assumes. We mechanize the transition system and the invariant.

2. **The soundness proof has a fully LCA-legal defeater** (§4.2): a reachable configuration of a legal add-wins MRDT — its final merge sitting at an honest LCA — on which every bottom-up peel of the paper's derivation rules is stuck. The meta-theorem must be rebuilt, not patched.

3. **The lattice contract does not transfer; a delta contract does** (§6): the lattice laws are equations over *free* tuples — the wrong instances for an LCA-aware merge. Side-idempotence $\mathsf{mergeL}(l, a, a) = a$ with l free fails for the counter $(2a - l)$, but its only DAG-honest instance has $l = a$ (the LCA of a version with itself *is* that version), where it holds trivially; and when two *distinct* histories reach equal side states, $2a - l$ is the *correct* merge — both branches' increments count — not an idempotence failure. The induced orders degenerate, and the honest associativity law has *four distinct LCAs* and holds only on feasible tuples. The moral: the algebra of an LCA-aware merge is an algebra of *causal histories*, not of concrete states. What replaces the lattice laws is a two-law *delta contract* — redistribution identities in which the LCA slot does, by arithmetic, the job idempotence does order-theoretically for CRDTs.

4. **Eight VCs suffice** (§7–§8): three relational VCs on rc (one of them carrying a different-replica guard — same-replica pairs are ordered by program order, and demanding rc cover them is unsatisfiable for real data types), merge symmetry, three feasibility-conditioned delta laws, and one contextual *causal-delta* equation. The adequacy theorem is per *version*: every version in the store, LCAs included, linearizes.

5. **The VC set is validated on the production catalogue** (§9): eleven of the twelve MRDTs shipped in Sal — OR-set, an optimized OR-set, enable-wins flag, grow-only set and map, increment-only and PN counters, a tombstone RGA, Peritext, the multi-valued register and the add-wins priority queue — are discharged end-to-end, kernel-checked; the

twelfth, the tombstone-free RGA, goes through the conditioned framework. The sweep produces an empirical *class map* of which data types need which strength of contract, with two surprises: tombstones are a metatheoretic trivializer, and commutativity class and delta class are independent axes.

6. **The eight VCs are one instance of a closure-indexed contract** (§15): a mechanized investigation of the note's own open questions establishes that the "second route" asymmetry (§8) is not a defect to be reunified but a *closure parameter* the data type declares; a single soundness bridge covers both strengths, and *all nine* production discharges route through it unchanged. The same investigation refutes the obvious attacks on the two remaining frontiers — it defeats even the commuting 2-OP rule on the defeater, pins the feasible-update-layer feasibility notion, and characterizes exactly why the tombstone-free RGA resists (it converges, but through semantic commutation the pointwise bubble cannot see). Pushed to its end, the investigation *hosts* the tombstone-free RGA: the swap architecture itself is refuted at merge unions, and a conditioned metatheory — an observational quotient, a canonical-witness discipline per version, and a generation discipline derived from born accuracy — proves it RA-linearizable up to observational equivalence, end-to-end under honest delivery. All kernel-checked.

Everything stated as a theorem below is mechanized with zero `sorry`s; §16 maps statements to Lean names and files (foundations under `Framework/`, metatheorems under `Metatheory/`, negative results under `Refutations/`, the per-RDT instances under `MRDT_Instances/`, the investigation record under `Development/`; each cited theorem kernel-checked).

2 Mergeable replicated data types, by definition and example

Data type and implementation. A replicated data type of *type* $\tau = \langle O_\tau, Q_\tau, Val_\tau\rangle$ exposes a set of update operations O_τ, a set of query operations Q_τ, and a domain of query return values Val_τ. An *MRDT implementation* of τ is a tuple

$$\mathcal{D} \;=\; \langle \Sigma,\ \sigma_0,\ \mathsf{do},\ \mathsf{mergeL},\ \mathsf{query},\ \mathsf{rc}\rangle,$$

where Σ is the space of replica states with initial state σ_0; $\mathsf{do} : \Sigma \times \mathcal{E} \to \Sigma$ applies an *event* — an update-operation instance $e = (t, r, o)$ carrying a globally unique timestamp t, its origin replica r, and the operation $o \in O_\tau$; $\mathsf{mergeL} : \Sigma^3 \to \Sigma$ is the three-way merge, whose first argument is the state at the lowest common ancestor of the two versions being merged; $\mathsf{query} : \Sigma \times Q_\tau \to Val_\tau$ reads a state; and $\mathsf{rc} \subseteq O_\tau \times O_\tau$ is the *conflict-resolution* relation, consulted to order concurrent non-commuting operations (it is the data-type designer's declaration of who wins a race). Queries never change state and play no role in the metatheory; the theory below reads the implementation as $\langle \Sigma, \sigma_0, \mathsf{do}, \mathsf{mergeL}, \mathsf{rc}\rangle$.

Describing an MRDT: the counter. To describe an MRDT one gives the five components and, when operations can race, says who wins. The increment-only counter of Figure 1 is the smallest complete example:

$$\Sigma = \mathbb{Z}, \qquad \sigma_0 = 0, \qquad \mathsf{do}(\sigma, (t, r, \mathsf{inc})) = \sigma + 1, \qquad \mathsf{mergeL}(l, a, b) = a + b - l, \qquad \mathsf{rc} = \emptyset.$$

All operations commute, so rc is empty — yet the merge genuinely uses its LCA argument: $a + b$ alone would double-count the history shared by the two branches, and the $-\,l$ subtracts it exactly once. This "read the shared past off the LCA" pattern recurs throughout the catalogue.

A data type with races: the OR-set. The add-wins observed-remove set stores tagged elements: add inserts a tag unique to the adding event, rem deletes exactly the tags it has *seen*. Sequentially, a remove after an add deletes it; concurrently the designer declares the add the winner: $\mathsf{rem} \xrightarrow{\mathsf{rc}} \mathsf{add}$. The merge keeps a tag iff both sides have it (nobody deleted it) or some side added it since the fork:

$$\mathsf{mergeL}(l, a, b) \;=\; (a \cap b) \cup (a \setminus l) \cup (b \setminus l).$$

The LCA here plays the role a tombstone set plays for the CRDT OR-set: deletion is absence *relative to* l, and the state needs no graveyard.

The generalized signature. Both examples' operations make sense on *every* state: any state can be incremented, any tag can be added. Such data types are *flat*, and for them the metatheory of §3–§9 quantifies its VCs over all states. But a data type whose operations carry *preconditions* — insert under a node that must exist, delete a node that must be live — breaks that quantification: its commutation equations are simply false at nonsense states. The full signature therefore extends the tuple with three declarations (`ConditionedMRDTSig`; formalized in §11):

$$\mathcal{D} \;=\; \langle \Sigma, \sigma_0, \mathsf{do}, \mathsf{mergeL}, \mathsf{rc},\ \mathsf{Inv},\ \mathsf{app} \rangle \quad \text{together with} \quad \approx\, \subseteq \Sigma \times \Sigma,$$

where Inv is a state invariant (a shape over-approximation of reachability), app an applicability predicate (when an event is sensible at a state), and $\approx$ an observational equivalence (what queries can distinguish; representation residue is quotiented away). Commutation is then required only at Inv-states where both events are applicable. Flat data types take $\mathsf{Inv} = \mathsf{app} = \top$ and $\approx\, = \,=$, collapsing the general signature to the plain one — one framework, of which §3–§9 is the flat instance and §10 onwards the general case.

The running hard example: the tombstone-free RGA. The Replicated Growable Array is the list CRDT/MRDT underlying collaborative text editing: characters are nodes identified by timestamps, each inserted *after* an anchor node. Production RGAs keep deleted nodes as *tombstones* because later concurrent inserts may still anchor on them. The *tombstone-free* RGA deletes physically, and it is describable in the general signature:

- Σ: finite forests — partial maps from node ids to (element, anchor id), the root id 0 never stored; σ_0 the empty forest.
- Operations: $\mathsf{Ins}(e, p, a)$ inserts a fresh node with element e under anchor a, and $\mathsf{Del}(p, x)$ deletes node x; both carry a *recorded path* p — the anchor's (resp. target's) ancestor chain as the issuing replica saw it. do resolves the first live entry of $a :: p$ and attaches there; a delete removes x physically and *rehomes* x's children to x's nearest live ancestor.
- $\mathsf{mergeL}(l, a, b)$: OR-set survivorship on node ids (present in both sides, or born since the fork on one), each survivor re-anchored by climbing its recorded chain to the nearest survivor.
- rc: empty in the directional sense (`Either`) — every pair commutes, but only *conditionally*.
- Inv: well-formed forest (every stored anchor live or root), id-monotone anchors ($\mathsf{anc}(x) < x$), root unstored. app: *accurate* (the recorded path is the target's true live ancestor chain in the current state) and *fresh* (the id is unused). $\approx$: equal live forests — same nodes, elements and anchors.

A three-node example shows why every piece is needed. Start from the chain $0 \leftarrow 1 \leftarrow 2$ (node 1 under the root, node 2 under 1). $\mathsf{Del}([0], 1)$ removes node 1 and rehomes 2 to the root. A *concurrent* event $(3, r, \mathsf{Ins}(e, [0], 1))$ — generated before the delete was seen, anchored on 1, whose recorded chain 1 :: [0] is 1's live ancestor chain at generation — arrives after it: the anchor is gone, and the recorded chain is exactly what lets do resolve the insert to 1's parent 0, where the delete rehomed 1's children. Commutation of this pair over *all* states is false (the recorded path can lie about a nonsense state); at reachable, accurate states it holds. This data type is the reason the general signature exists, and §10–§14 prove it RA-linearizable up to $\approx$, end-to-end.

3 The ternary setting

This section fixes the model: the data-type signature, the replicated store and its transitions, and the specification the metatheorem must deliver. The definitions are the paper's, with one correction flagged where it occurs.

Signature. An MRDT is $\langle \Sigma, \sigma_0, \mathsf{do}, \mathsf{mergeL}, \mathsf{rc} \rangle$ as in §2, read flat for now ($\mathsf{Inv} = \mathsf{app} = \top$, $\approx\ =\ =$; the general case resumes in §10). We write $e(\sigma)$ for $\mathsf{do}(\sigma, e)$, and $e_1 \rightleftarrows e_2$ when $e_1(e_2(\sigma)) = e_2(e_1(\sigma))$ for every σ.

Executions. A configuration carries a *store*: a DAG of versions, each holding a state and the set of events that produced it, with per-replica heads. Transitions fork a replica, apply a fresh event at a head, query, or **merge**: pick heads v_1, v_2, find their LCA v_ℓ in the DAG, and install a new version with state $\mathsf{mergeL}(\sigma_{v_\ell}, \sigma_{v_1}, \sigma_{v_2})$ and event set $L(v_1) \cup L(v_2)$. Visibility $\xrightarrow{\mathsf{vis}}$ records what each event's issuing replica had seen.

RA-linearizability, per version. Write $e_1 \parallel e_2$ when neither $e_1 \xrightarrow{\mathsf{vis}} e_2$ nor $e_2 \xrightarrow{\mathsf{vis}} e_1$ (concurrency).

Definition 3.1 (linearization order, set-relative)**.** *For an event set E, the order lo^E puts e_1 before e_2 iff*

$$\big(e_1 \xrightarrow{\mathsf{vis}} e_2 \ \wedge\ e_1 \not\rightleftarrows e_2\big) \ \ \vee\ \ \big(e_1 \parallel e_2 \ \wedge\ e_1 \xrightarrow{\mathsf{rc}} e_2 \ \wedge\ \neg\,\exists e_3 \in E.\ \ e_2 \xrightarrow{\mathsf{vis}} e_3 \wedge e_2 \not\rightleftarrows e_3\big) :$$

a visible non-commuting pair is ordered by visibility; a concurrent pair whose conflict rc *resolves is ordered by* rc *— unless e_2 is already absorbed (overwritten) by a later non-commuting event of E. The paper's order* lo *is the variant whose absorber clause ranges over the whole configuration; since a* lo*-edge asserts no absorber anywhere, $\mathsf{lo} \subseteq \mathsf{lo}^E$ pointwise, for every E.*

Definition 3.2 (RA-linearizability)**.** *A list $\pi = e^1 e^2 \cdots e^n$ witnesses a version holding state s and event set E when (i) π enumerates E without repetition; (ii) π extends* lo*: $i < j$ implies $\neg(e^j\ \mathsf{lo}\ e^i)$; and (iii) π folds to the state: $e^n(\cdots e^2(e^1(\sigma_0))\cdots) = s$. A configuration is RA-linearizable when every version in the store — not only the replica heads; LCAs are historical versions and the merge reads them — admits a witness. In the general signature the fold clause (iii) is read up to the data type's observational equivalence, $e^n(\cdots e^1(\sigma_0)\cdots) \approx s$ — the same definition with its equality parameter generalized; flat data types instantiate $\approx\ =\ =$ and this section's form is exactly that instance.* `IsRALinearizable3`, `IsRALinearizable3Eq`

Why a set-relative order? One definitional correction is forced at the outset: for every internal use, the absorber clause must range over the event set of *the version being linearized*, not over the whole configuration — an event can gain absorbers from another branch, silently cancelling an order edge inside a version whose own state depends on it (the defeater of §4.2 realizes exactly this). We therefore construct witnesses respecting lo^E throughout; since $\mathsf{lo} \subseteq \mathsf{lo}^E$, every such witness also extends the paper's order, and Definition 3.2 is delivered exactly as the paper states it.

Running examples. Three data types exercise every corner of the theory:

- the **counter**: $\Sigma = \mathbb{Z}$, $\mathsf{inc}(\sigma) = \sigma + 1$, $\mathsf{mergeL}(l, a, b) = a + b - l$; all operations commute, but the merge genuinely uses its LCA;
- the **OR-set** (tagged add-wins set): elements are tagged by the adding event's timestamp; rem deletes the tags it has seen; concurrently, $\mathsf{rem} \xrightarrow{\mathsf{rc}} \mathsf{add}$ resolves in favour of the add. The merge keeps an element iff *both* sides have it or *some* side added it since the fork:

$$\mathsf{mergeL}(l, a, b) \;=\; (a \cap b) \,\cup\, (a \setminus l) \,\cup\, (b \setminus l)$$

  — pointwise, "if the tag is in l, require it in both branches (nobody deleted it); otherwise one branch suffices (somebody added it)." Here the LCA acts as a *tombstone set*: deletion is represented by *absence relative to* l, so the state itself needs no graveyard;
- the **enable-wins flag**: per-replica pairs (counter, flag); enabling bumps your own counter and sets your flag; disabling clears all flags; the merge compares each replica's counter against the LCA's to decide whether an enable "happened since the fork" on that replica. Its merge *compares* counters rather than accumulating sets — and that difference will be visible in the metatheory (§9).

4 Three facts the metatheory must respect

Before building anything, we record two boundary facts about the paper's development: its store-level LCA lemma is true but under-proved, and its soundness induction is refuted by a fully legal execution. Together they fix the shape of the corrected metatheory — resting on a mechanized store invariant (§4.1) and built around the join rather than the sides (§4.2).

4.1 The LCA lemma is a reachability invariant — with a gap in the published proof

Everything above silently used:

Lemma 4.1 (LCA lemma)**.** *In every reachable configuration, if v_ℓ is an LCA of versions v_1, v_2 in the store's DAG, then $L(v_\ell) = L(v_1) \cap L(v_2)$.*

This is what lets merge VCs be stated over *event sets*: the state the merge receives as l is the canonical state of exactly the intersection of the two sides' histories. The lemma is true — but it is a *global induction over the reachable store*, not a local fact. Its mechanized proof threads a store invariant (all parents allocated; event sets monotone along edges; and an *origin* invariant — each event appears first at a unique generator version) through every transition; the published proof asserts the generator-uniqueness step without the induction that sustains it. An erratum-sized gap: the statement survives, the proof as written does not. (A related boundary fact: *criss-cross* merges can defeat the *existence* of an LCA — they gate merge-enabledness, not the lemma. Open Question 15.6 takes up the extension.)

4.2 A fully LCA-legal execution defeats the soundness proof

The generic half of the paper's contract is a *bottom-up linearization* theorem: given linearization witnesses for the two sides of a merge, construct one for the merged version by repeatedly *peeling* a final event through mergeL using the catalogue's interchange VCs. One might hope the version DAG's discipline — every merge sitting at an honest LCA — keeps the peel supplied with peelable events. It does not: one staging replica suffices to realize the intersection of two histories as an honest version, and the merge induction is stuck.

Example 4.2 (the defeater)**.** *The data type is a one-key add-wins skeleton with tagged adds: $\sigma = (A, D) \in 2^{\mathbb{T}} \times 2^{\mathbb{T}}$ with live set $A \setminus D$; add_t inserts its tag into A; rem kills everything it currently sees ($D \coloneqq A \cup D$ — the state-dependence is what makes $\mathsf{add} \not\rightleftarrows \mathsf{rem}$); concurrently $\mathsf{rem} \xrightarrow{\mathsf{rc}} \mathsf{add}$ (add wins); the merge is componentwise union, ignoring its LCA argument. A legal MRDT. Replica p adds (A_p), replica q adds (A_q). A staging replica s merges both one-add versions, creating v_s with $L(v_s) = \{A_p, A_q\}$. Now p removes (R_p, seeing only A_p) and then merges with v_s; symmetrically q removes (R_q) and merges with v_s. Figure 2 shows the DAG. The final merge of the two heads finds $\mathrm{LCA} = v_s$, and $L(v_s) = \{A_p, A_q\} = L(head_p) \cap L(head_q)$ — the configuration is fully legal. Now count last events. p's head lives on exactly A_q's tag, so a state-correct witness must run R_p before A_q, and neither R_p nor A_p can come last: every linearization of p's head ends in A_q. Symmetrically, every linearization of q's head ends in A_p. Yet in the merged version every tag is dead, so its linearizations must end in one of the removes. The events the induction must peel are buried in the very sides that own them; no ordering of the paper's bottom-up rules applies.*

The two removes commute ($\mathsf{rem} \not\rightleftarrows \mathsf{add}$ but $\mathsf{rem} \rightleftarrows \mathsf{rem}$), which invites a particular rescue attempt: the paper's BottomUp-2-OP rule (`inter_lca_2op`) admits the commuting case and could, one hopes, peel a remove last. It cannot, for a reason that is a one-line state fact. To peel o last from a side, that side's state must be an o-image — of the form $\mathsf{do}(\sigma', o)$ with o outermost. But in the add-wins skeleton *every* rem-output has empty live set ($\mathsf{rem}(A, D) = (A, A \cup D)$), whereas both heads carry a live tag (head_p lives on A_q, head_q on A_p — the opposite add, merged in *after* the local remove). So neither head is a rem-image, and no bottom-up rule can extract a remove from it. The tempting linearization $[A_p, R_p, A_q, R_q]$ is a valid witness of the *merged* version, but its restriction to head_q folds to the all-dead state, not head_q's actual state (live A_p): it is not assembled from a state-correct side witness, which is exactly what the induction requires. All three facts are mechanized (`InterLca2op_Defeater_Arbiter`: `awset_rem_output_empty`, `no_inter_lca_2op_rem_peel_of_defeater`, `crack1_witness`).

The consequence for the design: the metatheorem cannot be patched peel-by-peel — it must be rebuilt on canonical states and a Join Lemma, with merge-side VCs that are statements *about the join*, not about how either side's state factors.

5 Canonical states and the ternary Join Lemma

We now rebuild — new proof architecture, and with it a *new set of verification conditions* to replace the paper's catalogue. The lesson of the defeater is architectural: witnesses for a merged version cannot be manufactured out of witnesses for its sides, so the corrected metatheory must not traffic in witness lists at all. The rebuild has two layers. The *update layer* makes states, not sequences, the objects of the induction: it assigns to every event set E a single well-defined state, its *canonical state* (Definition 5.2 below). The *merge layer* is then one equation between

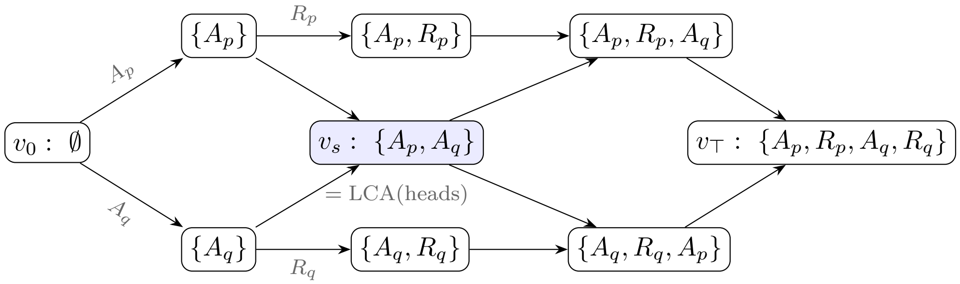


Figure 2: The defeater (Example 4.2). Nodes are versions labelled by their event sets; the shaded version v_s, created by a staging replica, realizes $L(\text{head}_p) \cap L(\text{head}_q)$ as an honest store version, so the final merge is LCA-legal. The removes R_p, R_q — the only events a bottom-up derivation may peel last — are buried mid-history on their own sides.

canonical states — the ternary *Join Lemma* — and establishing it from per-data-type VCs becomes the entire burden of the metatheorem (§6–§7).

The update layer comes first, and a structural observation makes it cheap: this half of the corrected theory never mentions the merge. Its one theorem is stated over the update signature $\langle \Sigma, \sigma_0, \mathsf{do}, \mathsf{rc} \rangle$ alone:

Theorem 5.1 (convergence)**.** *Assume the update-layer VCs (items 1–3 of §7). Then any two* lo^E*-respecting enumerations of a finite event set* E *fold from* σ_0 *to the same state.*

This is where the set-relative reading of §3 earns its keep: with the paper's configuration-wide order in place of lo^E, the theorem is *false* — the defeater's heads are already counterexamples, since two order-respecting replays of the three events of head_p reach different states. Convergence licenses a definition:

Definition 5.2 (canonical state)**.** *For a finite event set* E*, the canonical state* $\sigma(E)$ *is the state obtained by folding any* lo^E*-respecting enumeration of* E *from* σ_0*. At least one respecting enumeration exists because* lo^E *is acyclic (VC 2); the result is independent of the choice by Theorem 5.1; and* $\sigma(\emptyset) = \sigma_0$*. We write* $\sigma(E)$ *for the canonical state of an event set, and* σ_v *for the state a version* v *happens to hold in the store — proving that the two coincide is precisely the metatheorem's job.*

With canonical states in hand, witness lists disappear as promised: a version linearizes iff $\sigma_v = \sigma(L(v))$ — the witness, when one is owed, is any respecting enumeration of $L(v)$. The induction can therefore maintain a state-level invariant ("every version holds σ of its events") instead of constructing sequences. The one genuinely ternary statement is what the merge must produce:

Definition 5.3 (ternary Join Lemma)**.** *For backward-closed event sets* E_1, E_2 *of a reachable configuration,*

$$\sigma(E_1 \cup E_2) \;=\; \mathsf{mergeL}\big(\sigma(E_1 \cap E_2),\ \sigma(E_1),\ \sigma(E_2)\big).$$

By Lemma 4.1 the first argument is exactly the state the store hands the merge. The adequacy proof (§8) is then an induction over the transition system maintaining "every version holds the canonical state of its event set"; the merge case *is* the Join Lemma. What remains is the question this note exists to answer: which per-data-type VCs make the Join Lemma true?

6 Why the contract is a delta contract

Where should the laws that prove the Join Lemma come from? The obvious supplier is the CRDT tradition. For state-based CRDTs — binary merge, no LCA — merge-coherence is lattice-flavoured: associativity, idempotence, inflation of updates. None of the three transfers to the ternary merge: their free-tuple forms are the wrong equations for an LCA-aware merge, and their honest instances are too weak to power a lattice-style proof. This section works through why, because the shape of what fails to transfer reveals the contract that succeeds.

Idempotence: the free-tuple form is the wrong equation. The *diagonal* law $\mathsf{mergeL}(a, a, a) = a$ — the paper's own MERGEIDEMPOTENCE — holds (for the counter: $a + a - a = a$), but it is not the law a lattice-style proof consumes. The binary architecture uses idempotence to absorb a duplicated *side*, $\mathsf{merge}(a, a) = a$, with nothing tying the two copies to a common past; the ternary analogue would be $\mathsf{mergeL}(l, a, a) = a$ with l *free*. But quantifying l freely asks the merge about tuples the version DAG never produces. Merging a version with *itself* has $l = a$ — the LCA of a and a is a — where the law collapses to the diagonal and holds. And when two *distinct* histories happen to reach equal side states, feasibility puts l strictly below them and the counter's $\mathsf{mergeL}(l, a, a) = 2a - l$ is the *correct* answer: both branches' increments count. So there is no idempotence *failure* — there is no meaningful side-idempotence *law*: its honest instances are the trivial diagonal, and demanding it unconditionally would exclude precisely the LCA-sensitive data types the ternary theorem exists for.

The general principle, which the rest of this section instantiates: the algebraic properties of an LCA-aware merge are properties of the *causal history*, not of the concrete state. Two sides merge idempotently when their *histories* coincide — which is what forces $l = a$ — not when their states happen to be equal; equal states with different histories rightly merge differently. The lattice laws go wrong exactly by quantifying over states while forgetting the histories that produced them, and every law that survives below is stated over history-indexed data: canonical states $\sigma(E)$, feasible tuples (the LCA slot carrying the intersection history), downset deltas.

No usable order. The lattice-style workhorse order $x \sqsubseteq y := \mathsf{merge}(x, y) = y$ has ternary candidates $x \sqsubseteq_l y := \mathsf{mergeL}(l, x, y) = y$, but for the counter this relation degenerates to $x = l$; no antisymmetry survives to power order-theoretic reasoning. The single contextual VC below is accordingly an *equation*, not an inequality.

Associativity is the wrong target. The natural guess — $\mathsf{mergeL}(l, \mathsf{mergeL}(l, a, b), c) = \mathsf{mergeL}(l, a, \mathsf{mergeL}(l, b, c))$ — assumes one LCA for all three merges. In an execution, re-associating a three-way reconciliation involves *four* distinct LCAs:

$$\mathsf{mergeL}\big(l_{(ab)c},\ \mathsf{mergeL}(l_{ab}, a, b),\ c\big) \overset{?}{=} \mathsf{mergeL}\big(l_{a(bc)},\ a,\ \mathsf{mergeL}(l_{bc}, b, c)\big),$$

with $l_{ab} = \sigma(E_a \cap E_b)$, $l_{(ab)c} = \sigma((E_a \cup E_b) \cap E_c)$, and so on. For the counter — whose canonical states are additive measures over event sets — the two sides come to $a + b + c$ minus the two LCA measures, and they agree because

$$(E_a \cup E_b) \cap E_c = (E_a \cap E_c) \cup (E_b \cap E_c) \quad \Longrightarrow \quad \text{both LCA-sums} = |E_{ab}| + |E_{ac}| + |E_{bc}| - |E_{abc}|$$

by inclusion–exclusion. The law holds, but only *through the set-algebra of intersections*: over four unconstrained states it is false. Honest associativity is inherently a feasible-tuple fact —

and, the mechanization's pleasant surprise, **the corrected induction never needs it**. Every merge the induction forms already sits at its honest LCA; the only laws consumed are two *redistribution* identities that equate two honestly-formed merges:

Definition 6.1 (the delta contract)**.**

$$\begin{aligned} \textit{LOCAL-REDISTRIBUTE}: \quad & \mathsf{mergeL}(l,\, \mathsf{mergeL}(m, x, c),\, y) \;=\; \mathsf{mergeL}(m,\, \mathsf{mergeL}(l, x, y),\, c), \\ \textit{REDISTRIBUTE}: \quad & \mathsf{mergeL}\big(\mathsf{mergeL}(m, x_0, c),\, \mathsf{mergeL}(m, x_1, c),\, \mathsf{mergeL}(m, x_2, c)\big) \\ & \qquad = \; \mathsf{mergeL}\big(m,\, \mathsf{mergeL}(x_0, x_1, x_2),\, c\big). \end{aligned}$$

Read c as a *delta relative to* m (in the induction, c will be "apply event e to its causal past"). LOCAL-REDISTRIBUTE says a delta application commutes past an enclosing merge through one branch slot. REDISTRIBUTE says a delta carried by *all three* components extracts *once* — the LCA slot cancels the duplicates by arithmetic. This is the delta contract's quiet coup: cancelling the duplicated shared contribution is exactly the job idempotence performs in lattice-based convergence arguments, now done without any idempotence — which is why the counter sails through (REDISTRIBUTE for the counter is the identity $\sum x_i + 3c - 3m - (x_0 + c - m) - \cdots$ collapsing on both sides).

Where the contract holds unconditionally — and where it cannot. The two laws hold *on all states* exactly for the group-like class (the counter) and the lattice-like, LCA-blind class (grow-only structures). They are *unconditionally false* for the OR-set: take an element tag present in c and l but in neither branch — the two sides of LOCAL-REDISTRIBUTE then disagree about whether the LCA's copy survives. But such a tuple can never arise: a tag in a canonical LCA state is in *both* branches unless removed, and a removal is itself an event of the branch. The failing tuples are *infeasible*, and this is a theorem-grade phenomenon, not an inconvenience:

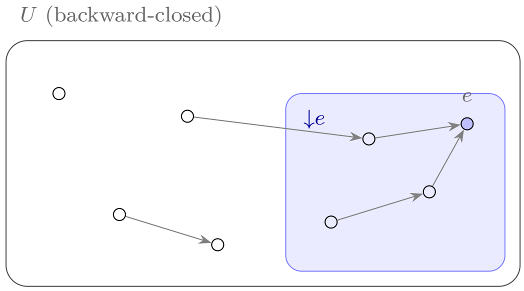
Finding. *For LCA-sensitive MRDTs beyond the group $\oplus$ lattice classes, the merge-coherence laws are irreducibly feasibility-conditioned: true on canonical tuples at honest LCAs, false in general. Quantifying merge VCs over all states — the published catalogue's style — is not merely inconvenient for such data types; for several laws it is wrong in principle. (The enable-wins flag even falsifies the unit law $\mathsf{mergeL}(\sigma_0, \sigma_0, s) = s$ on an infeasible state with a set flag and a zero counter.)*

Accordingly the delta laws enter the final contract in feasibility-conditioned form: quantified over canonical states of backward-closed event sets, with every mergeL node at its honest LCA. The unconditional form remains available as a one-line *on-ramp* (it trivially implies the conditioned form) for the group and lattice classes.

7 The eight verification conditions

The update-side obligations of §5 and the merge-side laws of §6 now assemble into the full contract — a *new* VC set, replacing the paper's catalogue of twenty-four. For an MRDT $\langle \Sigma, \sigma_0, \mathsf{do}, \mathsf{mergeL}, \mathsf{rc} \rangle$, the contract is:

Update layer (unconditional, SMT-shaped).



$A = \sigma(U \setminus \{e\})$ $\quad$ $B = \sigma(\downarrow e \setminus \{e\})$ $\quad$ CDVC3 : $\mathsf{mergeL}(B,\ A,\ e(B)) \ = \ e(A)$

Figure 3: The causal-delta equation. The shaded region is e's causal past $\downarrow e$ (what e saw); e is lo^U-maximal. e's effect on the whole world A equals: compute e's effect on its own causal past B, then merge that delta into A *over LCA* B. The merge sits at its honest LCA: $(U \setminus \{e\}) \cap \downarrow e = \downarrow e \setminus \{e\}$. Concurrent context that e never observed cannot amplify what e does.

1. **rc-non-comm (guarded)** — for distinct events *from different replicas*: they fail to commute iff rc orders them in exactly one direction. *The conflict table is the commutativity table, with a direction.* The different-replica guard is a semantic necessity, not a convenience: same-replica events are totally ordered by program order, so they are never concurrent and there is no conflict for rc to resolve. An unguarded VC would instead demand that rc order *every* non-commuting pair — unsatisfiable for real data types: the *optimized OR-set*'s same-replica same-element adds do not commute (the newer add evicts the older tag), yet they can never race, and its rc rightly ignores them. (The paper's F* artifact carries exactly this guard in its interface — `App_mrdt.fsti`: `requires distinct_ops o1 o2 /\ get_rid o1 <> get_rid o2`.)

2. **no-rc-chain** — conflict priorities do not chain: never $o_1 \xrightarrow{\mathsf{rc}} o_2$ and $o_2 \xrightarrow{\mathsf{rc}} o_3$. This is what makes the linearization order acyclic, so maximal events exist — the entire peel-and-recurse architecture stands on it.

3. **cond-comm** — a resolved conflict is invisible once an absorber runs: swapping an rc-ordered concurrent pair deep in a fold changes nothing for any later non-commuting event, across arbitrary intervening events. This is the engine of convergence.

Merge layer.

4. **merge symmetry** — $\mathsf{mergeL}(l, a, b) = \mathsf{mergeL}(l, b, a)$.

5. **feasible unit** — $\mathsf{mergeL}(\sigma_0, \sigma_0, \sigma(E)) = \sigma(E)$: merging over nothing is a no-op, on states that can arise.

6. **feasible local-redistribute** and

7. **feasible redistribute** — Definition 6.1, quantified over canonical tuples at honest LCAs.

8. **the causal-delta equation** (CDVC3) — for a lo^U-maximal event e of a backward-closed U, with $A = \sigma(U \setminus \{e\})$ and $B = \sigma(\downarrow e \setminus \{e\})$ the canonical state of e's own causal past:

$$\mathsf{mergeL}\big(B,\ A,\ e(B)\big) \ = \ e(A).$$

What an event does is determined by what it saw (Figure 3): applying e to the whole world equals computing e's delta on its causal past and merging it in over that past.

Items 1–4 are one-liners for every data type we have discharged; items 6–7 frequently hold unconditionally anyway (for the OR-set, REDISTRIBUTE is a pointwise Boolean tautology); essentially the entire per-data-type proof effort concentrates in item 8, whose discharge follows a uniform recipe: characterize $\sigma(E)$ (a "sandwich" invariant along canonical enumerations), then a *trichotomy* placing every event the maximal e interacts with either inside $\downarrow e$ or absorbed on a side that owns it.

Compare the published catalogue: 24 VCs per data type, plus five more the soundness proof uses silently. Seventeen of the 24 — the BottomUp induction-scheme families — exist to drive the broken induction and are consumed nowhere in the corrected one; the paper's MergeIdempotence is likewise never used. The seventeen were, however, doing honest work of a different kind: they are an *automation layer*, Skolemizing "holds on feasible states" into base-and-step equations an SMT solver can discharge. Rebuilding that layer, aimed at the corrected target (CDVC3 rather than BottomUp), is in our view the most valuable open engineering problem this work leaves (§15).

8 Adequacy

The contract delivers. Everything a data type owes is the eight equations of §7; everything else is proved once, for all data types, as part of the execution model.

Theorem 8.1 (soundness of the eight VCs)**.** *If an MRDT satisfies VCs 1–8, then every configuration reachable from the initial configuration in the ternary transition system is RA-linearizable, per version.* `ra_linearizable_of_core_feasible_cd3`

The proof shape: the update layer powers the set-relative convergence theorem, hence canonical states; the transition induction maintains "every store version is canonical for its event set" (fork and query are trivial; apply is the extend lemma); the merge case reduces, via Lemma 4.1, to the ternary Join Lemma, which is proved from VCs 4–8 by strong induction on $|E_1 \cup E_2|$: select a $\mathsf{lo}^{E_1 \cup E_2}$-maximal e, decompose each side containing e through $\downarrow e$ (LOCAL-REDISTRIBUTE at the honest LCA $\downarrow e \setminus \{e\}$), cancel the shared delta (REDISTRIBUTE), close the principal step with CDVC3, and recurse. The buried-event pathology of Example 4.2 dissolves structurally: no step ever asks how a *side's* state factors — every factoring happens through the join.

Everything else one might fear owing is proved once, for all data types, as an execution-model invariant: visibility transitivity, per-version backward closure, store well-formedness, and Lemma 4.1 itself. The data type owes exactly the eight equations.

A second route, and an honest asymmetry. One discharged data type does not fit the pattern above. The enable-wins flag's merge *compares counters* against the LCA, and for such merges the Join Lemma's hypotheses as stated (backward closure under non-commuting-visibility) are provably too weak: there is a closure-legal tuple — a live LCA enable, a dead post-LCA enable inflating one side's counter, the killing disable visible only to the other side — on which the production merge computes the wrong flag. Full *causal* closure (which the store invariant actually supplies) excludes it, and the flag is discharged through a full-closure variant of the join. The finding: **required closure strength is a function of the merge's shape** —

MRDT	contract tier	end-to-end theorem	σ-characterization (one line)
Grow-only set	unconditional	`GOSet_ra_linearizable3_eq`	union of adds
Grow-only map	unconditional	`GOMap_ra_linearizable3_eq`	union of (k, v) records
Incr.-only counter	unconditional	`IOC_ra_linearizable3_eq`	#events
PN-counter	unconditional	`PN_ra_linearizable3_eq`	#inc − #dec
RGA (tombstone)	unconditional	`RGAM_ra_linearizable3_eq`	grow-only nodes + tombstones
Peritext	unconditional	`Peritext_ra_linearizable3_eq`	grow-only mark records
OR-set	feasible	`ORSet_ra_linearizable3_eq`	tag live iff added, no vis-later `rem`
OR-set (efficient)	feasible	`ORSetE_ra_linearizable3_eq`	… or same-replica `add` later (eviction)
Enable-wins flag	full-closure	`EWFlag_ra_linearizable3_eq`	per key: ctr = #enables; flag iff unkilled enable
Multi-valued register	feasible	`MVR_ra_linearizable3_eq`	tagged writes × overwrite log; all-comm $\Rightarrow B{=}\sigma_0$
Add-wins priority queue	feasible	`AWPQ_ra_linearizable3_eq`	the OR-set pattern on A; grow-only I
RGA (tombstone-free)	*conditioned*	`rga_tombstone_free_ra_linearizable3_eq`	— RA-lin up to $\approx$, honest delivery (§15)

Table 1: The Sal production catalogue. Every instance concludes through the ONE generic conditioned metatheorem (§10–§14): the nine flat data types at the identity instantiation ($\approx = =$, their VC bundles feeding the flat collapse `FlatGeneric_Bridge.lean`, instance theorems in `MRDT_Instances/ (per-RDT)`), the tombstone-free RGA at full generality. The contract-tier column is each data type's Join-Lemma discharge content (`MRDT_Instances/`, one directory per RDT); the historical flat corollaries over the raw system remain there as internal steps.

commutation-closure suffices for set-shaped merges, counter-comparison merges need full causal closure. Reunifying the two routes was expected to be routine — "full closure only strengthens what a data type may assume." Mechanization shows it is not (§15, Open Question 15.3): the naive reunification — re-run the Join Lemma induction with full-closure hypotheses — is refuted, because no single-event (or block) peel preserves full closure. What survives is a *closure-indexed* contract in which the data type *declares* its closure strength; the soundness bridge is uniform across strengths, so no reunification into one strength is needed.

9 The catalogue, discharged

Table 1 is the empirical heart of the note: the VC set is not merely adequate in principle — the data types people actually ship satisfy it. Three lessons from the sweep:

Tombstones are a metatheoretic trivializer. The catalogue's celebrated hard cases — the RGA sequence type and the Peritext rich-text type — land in the *easiest* tier. The reason is almost embarrassing: with tombstones, deletion *adds* a record; every state component is grow-only; all operations commute; the merge is a blind join. All of the RGA's genuine difficulty lives in the *read-side* traversal (turning the node soup into a sequence), which RA-linearizability of the state never touches. By contrast the *tombstone-free* RGA — which physically deletes and repairs positions by climbing ancestor paths — is the one data type the eight VCs cannot host, for a reason worth stating precisely: its commutation VC is itself *reachability-conditioned* (two inserts commute only on well-formed states), and the update layer above, like the paper's, quantifies over all states. That was the question the tombstone-free RGA was built to force, and it is now answered — not by a conditioned update *layer* (every layer-level conditioning is refuted, see the feasible-update question in §15) but by a conditioned *metatheory* that abandons pointwise swaps altogether; the resulting end-to-end theorem is RA-linearizability up to observational equivalence under honest delivery.

Commutation class and delta class are independent axes. The multi-valued register is all-commuting, yet *not* in the unconditional tier: its merge is intersection-shaped, $(l \cap a \cap b) \cup (a \setminus l) \cup (b \setminus l)$, and fails the delta laws on infeasible tuples exactly as the OR-set does. The boundary between tiers is drawn by *how the merge uses its LCA*, not by whether operations commute. (The compensation: all-commuting makes every punctured causal past empty, $B = \sigma_0$, so the feasible discharge needs no trichotomy at all.)

The eight VCs certify real code on its historical fault line. The enable-wins flag's repository history contains a known-broken sibling: a global-counter variant whose compositional VC (`inter_right_1op`) fails on DAG-shaped executions, the very defect per-replica counters were introduced to fix. The mechanized σ-facts prove the production merge computes exactly union-liveness — *verified on precisely the corner where the sibling fails.* This is the methodology working as intended: the metatheory does not merely bless the catalogue, it explains *why* the fixed design is right.

10 The conditioned metatheorem

The eight VCs of §7 quantify over *all* states, and §9 showed that is right for flat data types and fatal for state-dependent ones: in the tombstone-free RGA of §2, two concurrent inserts commute only when each one's anchor is faithfully resolvable in the other's presence, and an insert under a since-deleted, physically-removed node is not even applicable. Written over all states, the commutation VC is simply false (a mechanized counterexample: `naive_general_swap_false`). The remedy is the general signature of §2: condition every VC on the state invariant Inv and the applicability predicate app, relax the witness fold of Definition 3.2 to the data type's observational equivalence $\approx$, and prove RA-linearizability from the conditioned VCs once and for all. The developer's burden is only to describe sensible operations; the metatheorem does the rest. This section states the conditioned VCs and the metatheorem with its pen-and-paper proof; §13 discharges the VCs for the tombstone-free RGA; §14 records how the mechanization actually went — including the machine-checked refutations that reshaped the route — and what the completed assembly proves.

11 The conditioned verification conditions

A datatype presents a signature $\langle \text{State}, \sigma_0, \mathsf{do}, \mathsf{merge}, \mathsf{Inv}, \mathsf{app} \rangle$ where Inv is a predicate on states and app a predicate on (operation, state) pairs (Lean: `ConditionedMRDTSig`, `Framework/MRDTSig.lean`). It must supply:

VC 1 (Discipline)**.** $\mathsf{Inv}\,\sigma_0$ *holds;* do *preserves* Inv *on applicable operations; and generation is disciplined: every operation is* app *and fresh at its birth state, with globally unique, monotone ids. (This is the execution model, Lean:* `ConditionedConfiguration`, `Framework/ConditionedExecutionModel.lea` *— kernel-clean, axiom-free.)*

VC 2 (Conditioned commutation — the swap witness)**.** *For concurrent operations a, b and any state σ with* $\mathsf{Inv}\,\sigma$ *at which both are applicable in the relevant reorder sense,* $\mathsf{do}\,(\mathsf{do}\,\sigma\,a)\,b \approx \mathsf{do}\,(\mathsf{do}\,\sigma\,b)\,a$.

VC 3 (Threadable reorder invariant). *An auxiliary invariant J on (state, pending-event) that (i) certifies applicability/the swap precondition of VC 2 at reorder states, (ii) holds at the enabling fold of each event, and (iii) is preserved by each reorder step. Only the enabled events — those whose causal past is folded — need be certified; that scope is exactly what a linearization repair ever transposes.*

VC 4 (Merge–linearization bridge — the conditioned Join Lemma). *For a reachable LCA state l and branches a, b obtained from l by disjoint concurrent event lists, $\mathsf{merge}\, l\, a\, b \approx \mathsf{fold}\, l\, \pi$ for a lo^E-respecting enumeration π of the branch events.*

For flat datatypes $\mathsf{Inv} = \mathsf{app} = \mathsf{true}$, VC 2 degenerates to unconditioned commutation, VC 3 to $J = \mathsf{true}$, and VC 4 to the ordinary Join Lemma — recovering the unconditioned methodology.

12 The metatheorem

Theorem 12.1 (Soundness — conditioned VCs $\Rightarrow$ RA-linearizability). *Any MRDT satisfying VCs 1–4 is RA-linearizable (Definition 3.2, read up to $\approx$).*

The proof is generic — it never inspects a particular datatype — and factors through two lemmas, each proved over the abstract signature with $\approx$ an arbitrary congruence for do and merge.

Remark 12.2 (Route actually taken). *The convergence lemma below is stated via the swap-based VC 3 for exposition. That route is machine-refuted for the tombstone-free RGA; convergence is mechanized by direct canonical-state characterization instead. See Section 14.*

Lemma 12.3 (Convergence). *VCs 1, 2, 3 imply that all lo^E-respecting admissible enumerations of a backward-closed reachable E fold from σ_0 to $\approx$-equal states; hence $\sigma^*(E)$ is well defined up to $\approx$.*

Proof. Any two lo^E-respecting enumerations of E are related by repeatedly transposing adjacent lo^E-incomparable events. Because lo^E is non-transitive, "respecting" is the elementwise `Pairwise` condition rather than sortedness, so this order-repair is more delicate than for a total order — it is exactly what the generic engine `eq_convergence` carries out, and we appeal to it rather than re-derive it. Each transposition preserves the fold up to $\approx$ by VC 2, provided the swap precondition holds at the transposition point; VC 3 supplies it along the whole enumeration (base at each event's enabling fold, preserved by each step, scoped to the enabled events — the only ones the repair transposes). The engine consumes only that $\approx$ is an equivalence, do is $\approx$-congruent, and a swap oracle of the shape VC 2+3 provides (Lean: `eq_convergence`, kernel-clean). ☐

Lemma 12.4 (Canonical adequacy). *VC 4 and Lemma 12.3 imply that every reachable version state is $\approx$ σ^* of its delivered event set.*

Proof. Induction over the version DAG. Initial: the empty fold. Update node: appends one delivered operation to a canonical fold (definitional). Merge node with LCA l and branches a, b: by hypothesis $a \approx \sigma^*(E_a)$ and $b \approx \sigma^*(E_b)$, and $l \approx \sigma^*(E_l)$. First rewrite a, b to literal canonical folds using $\approx$-congruence of merge (so VC 4, stated for branch folds, applies); then VC 4 gives $\mathsf{merge}\, l\, a\, b \approx \mathsf{fold}\, l\, \pi$ for a lo^E-respecting π of the branch events $E_a \cup E_b$. Now E_l

causally precedes those branch events (they were generated on states derived from l), so a lo^E-respecting enumeration of $E_l \cup E_a \cup E_b$ factors as (an lo^E-enumeration of E_l) followed by π; hence $\mathsf{fold}\, l\, \pi = \mathsf{fold}\, (\sigma^*(E_l))\, \pi \approx \sigma^*(E_l \cup E_a \cup E_b)$ by Lemma 12.3 (the engine is start-state generic). Backward closure keeps every enumeration admissible. □

Proof of Theorem 12.1. Immediate from Lemma 12.4: each version state $\approx \sigma^*(E_v)$, which is Definition 3.2. (Unconditioned template: `ra_linearizable3_of_joinC` plus the closure-indexed adequacy bridge, already kernel-clean with nine instances; the conditioned metatheorem generalizes it by carrying $\mathsf{Inv}/\mathsf{app}$ through the two lemmas.) □

Corollary 12.5 (Strong eventual consistency)**.** *Two versions with the same delivered set E have $\approx$ states: both $\approx \sigma^*(E)$ by Theorem 12.1, hence $\approx$ each other.*

13 The tombstone-free RGA discharges the conditioned VCs

The RGA is a forest of identified nodes; do inserts a fresh node under a resolved anchor or physically deletes a node and rehomes its children; merge keeps OR-set survivors and re-anchors each by climbing to the nearest surviving ancestor. $\mathsf{resolve}\, \sigma\, p$ returns the first live id on a recorded path p; $\mathsf{anc}\, \sigma\, x$ is x's current parent. Take $\mathsf{Inv} := \mathsf{wf} \wedge \mathsf{id\_mono} \wedge$ (root not reinserted) and $\mathsf{app} := \mathsf{accurate} \wedge \mathsf{fresh}$. This section presents the discharge as it was *first attacked*: swap-based, with the two hard VCs reduced to one faithfulness invariant. The attack succeeds — every lemma below is kernel-clean — but it is not what the final theorem stands on: a kernel-level dependency audit of the assembled capstone shows its proof term uses *none* of the swap machinery; of this section only the Key Lemma (Lemma 13.3) survives into the working proof. We keep the sketch because it is where the working technique's ingredients were found; Section 14 records what replaced it.

Definition 13.1 (Recorded-path faithfulness $\mathsf{RecPathFaithful}$)**.** *$\mathsf{RecPathFaithful}\, a\, \sigma$: the recorded path $\mathsf{rec}\, a$ is a live subchain of genuine ancestors of a's target in σ — true anc-ancestors, nearest first, consecutive pairs real parent edges, and exactly the chain live at a's generation. (This is `accurate/HistFaithful` promoted to a reachability invariant.)*

Lemma 13.2 ($\mathsf{RecPathFaithful}$ is a reachability invariant)**.** *$\mathsf{RecPathFaithful}\, a\, \sigma$ holds at every reachable σ in which a is applied.*

Proof. Induction on the operations building σ. *Birth:* do sets $\mathsf{rec}\, a = \mathsf{resolve}$-path at generation = the live-ancestor chain, so $\mathsf{RecPathFaithful}$ holds (`chainFaithful_of_histFaithful`). *Later* Ins *with fresh id f:* attaches f as a descendant — no reparenting, no liveness change for prior events' chains. f cannot appear spuriously in any $\mathsf{rec}\, a$: its members are older ids $< a.\mathsf{id} \leq$ every id extant at a's birth, whereas f exceeds all of them, so $f \notin \mathsf{rec}\, a$ definitionally. The resurrection/clash regime (`chainFaithful_not_preserved_under_clash_ins`) is therefore *unreachable* by freshness (`fresh_not_mem_recList`, `clash_recList_excluded`), not excluded by a side condition. *Later* $\mathsf{Del}\, x$*:* flips x's liveness only; $\mathsf{rec}\, a$'s entries stay genuine ancestors, one possibly now dead — still a live subchain (`chainFaithful_doDel_faithful`). □

Lemma 13.3 (Key Lemma: resolution over a live subchain)**.** *For any live subchain of genuine ancestors pre of x, in every reachable σ, $\mathsf{resolve}\, \sigma\, pre = \mathsf{anc}\, \sigma\, x$. (Lean: `subchain_resolve`, `RGA_SubchainResolve.lean`, kernel-clean.)*

Proof. resolve returns the first entry live in σ; each entry is a genuine ancestor, so this is the nearest live ancestor among pre. It is the nearest overall: no live ancestor $c \notin pre$ is nearer than that first-live entry — (a) if c was live at capture it was in the captured live chain, so $c \in pre$ (and any pre-entry nearer than the first-live one is dead in σ, hence not c); (b) if c was dead at capture it is still dead (physical, tombstone-free deletion is permanent); (c) an event inserted after capture attaches as a descendant, never a new ancestor of the pre-existing x. The mechanization absorbs these three cases into one preserved invariant $\mathsf{LiveChain}\,\sigma\,x\,pre := (\text{root unstored}) \wedge \mathsf{live}\,\sigma\,x \wedge \mathsf{IsAncPath}\,\sigma\,x\,(pre.\mathsf{filter\,live})$, which holds at capture, is preserved by each insert (`isAncPath_upd`) and delete (`isAncPath_surgery`, chain splicing across a deleted entry), and yields the conclusion. Case (b) is discharged by an explicit hypothesis $t \notin pre$ on inserts (`okStep`) — the mechanized form of "no resurrection," itself a consequence of monotone allocation, every pre-entry sitting below x's id. This replaces the earlier full-chain lemma `hres_of_lchain`, which failed on the proper subchains rehoming produces. □

Proposition 13.4 (RGA discharge)**.** *The RGA satisfies VCs 1–4. (Proof sketch below along the swap route; in the final mechanization the entire route — the swap VC, the* ChainFaithful *threading, the per-id* BranchInv *— is superseded by the canonical-state engine of Section 14, and only the Key Lemma survives; see the audit note there.)*

Proof. *VC 1* (discipline): Inv preservation and the generation model are `ConditionedConfiguration` instantiated at the RGA, with the RGA's do-invariance lemmas; `conditioned_premises` supplies the timestamp/freshness facts. *VC 2* (swap): `general_swap_bothFaithful` / `eqSwap_of_bothFaithful` — concurrent operations commute up to $\approx$ when both are *faithful* (neither need be exact/accurate), kernel-clean. *VC 3* (reorder invariant): $J :=$ ChainFaithful on the enabled scope, with steps `chainFaithful_incompStep`/`incompFold`; its *base* — each event faithful at its enabling fold — is exactly RecPathFaithful (Lemma 13.2) projected through the Key Lemma (Lemma 13.3): the residual goal $\mathsf{resolve}\,\sigma\,((\mathsf{rec}\,a) \setminus t') = anch'$ when an ancestor t' becomes leftmost live is the Key Lemma on the remaining subchain. *VC 4* (Join): per-id extensional match (`eq_merge2_of_branchInv2`); the anchor-coincidence clause is the single-sided I_4 (`eq_merge_single`, kernel-clean) and, cross-branch, the Key Lemma directly (`branchInv_doDel_crossBranch`, kernel-clean); order-independence is Lemma 12.3 at start state l. (In the final assembly this per-id BranchInv threading is itself superseded by the birth-anchor bridges of Section 14, which need no invariant threading at all.) Both hard VCs (3 base, 4 cross-branch) thus rest on RecPathFaithful and the Key Lemma — proved by construction from freshness and permanence. □

Corollary 13.5. *The tombstone-free RGA is RA-linearizable (Theorem 12.1 + Proposition 13.4), and strongly eventually consistent (Corollary 12.5). To our knowledge this is the first mechanized RA-linearizability / SEC result for an RGA with physical, tombstone-free deletion (cf. Gomes et al., OOPSLA'17, for the tombstoned RGA).*

Remark 13.6 (Other instances)**.** *The nine flat production MRDTs (counter, OR-set, MVR, …) take* $\mathsf{Inv} = \mathsf{app} = \mathsf{true}$*: VC 2 is their unconditioned commutation, VC 3 is* $J = \mathsf{true}$*, and VC 4 is the ordinary Join Lemma — already discharged (*`ra_linearizable3_of_joinC`*). They inherit RA-linearizability from the same Theorem 12.1. The RGA is the instance that exercises every conditioned mechanism; the framework supplies the generality, the RGA the novelty.*

14 How it was actually mechanized (and two dead ends)

The exposition above (Sections 7–**??**) is the *intended* route, and it correctly frames the contribution: conditioned VCs ⇒ RA-linearizability, RGA as instance. But the mechanization did *not* follow the swap-based convergence of VC 3. That route was tried and **machine-refuted twice**; convergence was then proved by a different technique. We record the actual route honestly — the refutations are themselves findings about why tombstone-free deletion is hard.

A postscript, established after the assembly was complete: a kernel-level dependency audit of the capstone's compiled proof term shows the subsumption is *total*. The final theorem's dependency cone contains no swap lemma, no ChainFaithful/RecPathFaithful threading, and no BranchInv — of the swap route only the Key Lemma remains load-bearing. The route's files stay proved and kernel-clean, retired under `Development/`; of the investigation's ∼17k lines, the capstone's dependency cone spans ∼8k. The lesson generalizes the refutations: for a datatype whose operations reference mutable structure, the set-indexed characterization is not just an alternative to linearize-and-swap — it makes the swap layer unnecessary even where the swaps are provable.

Two machine-checked refutations of the swap route. The reorder-invariant convergence (VC 3, "thread faithfulness through adjacent swaps") fails for the tombstone-free RGA because *rehoming makes an event's recorded chain time-relative*: correct at generation, wrong at reordered intermediate states.

- lo^E **is not transitive** (`RGA_DepComp_Gate`, kernel-clean refutation). Reordering dependencies-contiguous-first needs transitivity of the linearization order; a physical delete severs an operation's applicability-dependence on an ancestor ($b \to a \to o$ but $\neg(b \to o)$ after a's target is deleted). Witness: $l = 0 \leftarrow 1 \leftarrow 2 \leftarrow 3$, delete node 1, delete node 2.
- **In-place threading fails at the ancestor-insert step** (`RGA_SimulInduction2`, kernel-clean refutation). Even threading without reordering, an event's recorded chain can be *ahead* of the state (it anticipates a delete not yet folded), so ChainFaithful of the recorded chain is false at that intermediate fold although it heals later.

Both say the same thing: the linearize-and-swap method, which needs faithfulness at every *intermediate* reordered state, is the wrong vehicle. This is exactly what tombstones prevent (Gomes et al. keep deleted nodes so dependencies never rewire); the tombstone-free RGA rewires them.

The technique that works: direct canonical-state characterization. Instead of swaps, characterize the fold state as a pure function of the applied event *set* — the update analog of the merge bridge (which always worked). For a finite applied set F, define $\mathsf{survivors}\,F$ (OR-set survival) and $\mathsf{canonAnc}\,F\,k$ (climb k's recorded chain to the nearest survivor of F); let $\mathsf{CanonMatch}\,F\,s$ say s matches this per id.

Lemma 14.1 (Canonical fold)**.** *Along any* lo^E*-respecting enumeration* π*,* $\mathsf{CanonMatch}\ (\pi\text{-}applied)\ (\mathsf{fold}\ \sigma_0\ \pi)$ *(Lean:* `canon_fold`*, kernel-clean). Two folds of the same* E *then both match* $\mathsf{CanonMatch}\,E$*, hence are* $\approx$ *— convergence with no swaps and no intermediate-state faithfulness (*`RGA_update_convergence_canc` *discharged from the per-event generation discipline* GenDisc2C *in* `RGA_update_convergence_final`*).*

The invariant is over the *applied* set, so $\mathsf{canonAnc}\,F$ never anticipates a future delete — exactly why it succeeds where the swap route (which reasons about not-yet-applied events) fails.

The Key Lemma (Lemma 13.3) survives unchanged as the per-id climb-vs-resolve coincidence; RecPathFaithful is subsumed by CanonMatch's per-survivor `LiveChain`.

The metatheorem as a generic $\approx$-quotient functor. Theorem 12.1 is mechanized generically over `ConditionedMRDTSig` (`RA_linearizable_up_to_eq`, `GenericEqQuotient`, kernel-clean). It quotients the state by $\approx$ over the Inv-subtype and transfers the datatype's $\approx$-Join to the structural-= template `ra_linearizable3_of_joinC` by `Quotient.sound`; the readback is over *raw* fold, matching the RGA's convergence. The datatype supplies five VCs: EqEquiv, InvPres, CongVC (update/merge/query $\approx$-congruent on Inv), InvInvVC, and the $\approx$-Join EqJoinLemma3C. Two subtleties the mechanization forced, each machine-verified:

- **The merge $\approx$-congruence is false on the full state type** (`merge_eq_congr_l_fails`, axiom-free), true on the reachable subfamily ($\mathsf{wf} \wedge \mathsf{id\_mono} \wedge \mathsf{forest}$): the LCA-climb reads $\mathsf{anc}\, l$ off $\mathsf{domain}\, l$, where $\approx$ is silent. Hence the Inv-conditioning of CongVC is load-bearing, and the quotient is over the Inv-subtype (`merge_eq_congr_inv`).
- **Invariant preservation needs well-formedness, not applicability.** InvPres cannot demand unconditional preservation (do at a bad id breaks the invariant), nor full app (which the execution model does not guarantee — `Step3.apply` applies with only timestamp freshness). The honest condition is a datatype predicate WfOp (for the RGA: fresh id for Ins; $\mathsf{resolve}\, s\, p \neq x$ for $\mathsf{Del}\, p\, x$), preserved by the real do (`rgaInv_doOp_fresh`, accuracy *removed*) and holding along reachable folds (WfOpReachable). A totality guard on the lifted step is provably transparent on the execution domain ($\mathsf{applySeqW} = \mathsf{applySeq}$); the exposed theorem is over the real do.

The completed assembly. Everything after the quotient functor was, at the first writing of this note, "bounded assembly." Carrying it out surfaced three further machine-checked refutations that reshaped the target and the witness discipline; we record them because each is a fact about tombstone-free deletion, not about our proof.

- **At a merge union, applicability-tolerant witnesses do not exist.** An LCA-first enumeration pre-applies one branch's anchor-killing deletes before replaying the other branch's inserts, staling their recorded paths — so even *no-op-feasible* witnesses (each step applicable or a no-op) are unsatisfiable from the initial state (`RGA_HEnum_Refutation`). Worse, on a seven-event two-branch execution *no* born-applicable enumeration of the union replays both branches' recorded observations: rehoming makes the *order of concurrent deletes* observable in survivors' stored parents, and the branches observe it differently. Consequently the linearization target itself must be the *raw* do-fold, up to $\approx$ — any guarded or feasibility-filtered fold is unsatisfiable at merge unions (`IsRALinearizable3Eq`, `GoodConfig3H.lean`).
- **The witness discipline is the engine's own, plus payload honesty.** Each version carries, existentially, an enumeration satisfying the canonical engine's per-step discipline (`CanonFoldOK`: fresh nonzero ids, no id reuse, live recorded chains) *and* payload honesty (`HonestPayloads`: delete targets and recorded-chain entries are root-or-inserted). The second clause is forced by a counterexample: the fold discipline alone admits a dead-target delete (a no-op), and a later insert reusing that recorded dead id breaks the no-id-reuse clauses — honesty of *payloads* is set-level, hence permutation-invariant, and it is exactly what extension at a fresh apply needs (`rga_hHext_discharged`). The discipline *joins* at merges with the union witness being the LCA-then-delta concatenation $\rho_0 \mathbin{++} \pi_0$ itself: no

reordering, no swaps, anywhere (`rgaJoinH_of_canon`).

- **The generation discipline is derived, not assumed.** Every event is accurate at the fold of its *transitive dependencies* (`GenDisc2C`), provable from *born accuracy* — the op was generated against a causal fold of the events its replica had seen (`genDisc2C_of_born`). Merge canonicity then reduces to three leaves, all discharged: σ_0 id-monotonicity with *no fold induction* (the stored anchor is $\mathsf{canonAnc}$ of the recorded chain, whose entries are dependencies, which are Lamport-below — `rga_Hdec_discharged`); a per-id causal set-algebra whose only non-trivial clauses are dependency-fold provenance (`rga_hcaus_discharged`); and per-survivor *birth-anchor bridges* reducing the merged forest's anchors to record coherence at dependency folds, where the generation discipline makes every recorded chain live (`rga_hbridge_discharged`, `hin_of_genDisc`).

The residual, quantified once over the execution, is a single per-step assumption (`HonestDelivery`): *born accuracy* plus *born-applicable delivery.* It is genuinely irreducible — it is the generation discipline tombstone-freedom forces — and it is also all there is: Lamport clocks, timestamp uniqueness and visibility-support are structural fields of the configuration (dishonest-clock executions are unrepresentable); nonzero insert ids and nonzero delete targets follow from the delivered op's own wellformedness at a representative of its head class; nonzero delete *timestamps* are Lamport-derived from the target's insert. The end-to-end theorem (`rga_RA_linearizable_honest`; main-line entry point `rga_tombstone_free_ra_linearizable3_eq`, file `MRDT_Instances/RGA_TombstoneFree/RA_Lin`
closes Proposition 13.4 and the corollary unconditionally modulo that assumption.

15 Open questions

The mechanization settles what the paper claimed; it also sharpens what remains genuinely open. Six questions, roughly in order of theoretical depth:

Open Question 15.1 (CDVC3-derivability)**.** *Is* $\mathsf{CDVC3}$ *derivable from VCs 1–7 (plus the unconditional delta laws where they hold)? Specializing the merge to an LCA-blind lattice join — the CRDT case — the question closes exactly: there, the corresponding causal-delta bound is equivalent to the Join Lemma and no strictly weaker bridge VC exists, with both attack routes characterized — the positive induction is circular at two identified case shapes, and the countermodel space is narrowed to non-atomic lattice states (all of this mechanized, in* `Sal/CRDTs/Metatheory/`*). The ternary question is open.*

Open Question 15.2 (structure theorem)**.** *Empirically, the unconditional delta contract holds exactly on the group and lattice classes. Is every unconditional-*$\mathsf{DeltaVCs3}$ *merge a torsor-like group* $\otimes$ *lattice combination? (The naïve two-sorted umbrella* $\mathsf{mergeL}(l, a, b) = a \boxplus (b \boxminus l)$ *does not derive the contract uniformly, so the question is genuinely open.)*

Open Question 15.3 (route reunification — naive route refuted, closure-indexed resolution mechanized)**.** *The obvious reunification — restate the delta laws and* $\mathsf{CDVC3}$ *over full causal closure and re-run the Join Lemma induction — fails, and the failure is mechanized. The induction must peel an event that is both* lo^U*-maximal (placeable last) and vis-maximal (its removal keeps the sides fully closed); on a reachable four-event configuration these two maximal sets are disjoint, and the obstruction persists for every block peel, in both directions (*`JoinLemma3C: no_proper_back_block`*,* `no_proper_front_block`*). A wider induction class (fully-closed minus a* lo*-upward peel set) moves the obstruction but does not remove it: individually fully-closed sides admit no common witness set to decompose against (*`killTest_no_common_U`*). What does* *work needs no reunification: a* ***closure-indexed contract*** JoinLemma3C $\mathcal{C}$ *in which the data type declares its closure strength* $\mathcal{C}$*, with a single soundness bridge (*`ra_linearizable3_of_joinC`*): store versions are fully causally closed, so* JoinLemma3C $\mathcal{C}$ *applies for any* $\mathcal{C}$ *that full closure implies — both weak and full qualify, and the whole* `GoodConfig3` *induction is reused verbatim. The two routes of §8 are the two instances, and all nine production discharges route through the bridge unchanged (*`AllMRDTs_viaContract`*), reusing their existing VCs verbatim: eight at* (*weak*, $\top$) *— the OR-sets via the feasible triple, the six unconditional types via the delta on-ramp — and the enable-wins flag at* (*full*, $\top$)*; the disjunction is the contract, not a defect to be reunified. This is the "no new mathematics" the resolution promised — confirmed, with one correction: the bridge holds exactly for* $\mathcal{C}$ *weaker than full closure (a* $\mathcal{C}$ *stronger than full makes* JoinLemma3C *too weak to be adequate). Genuinely open: whether the single-strength contract (carrying the common witness set as data, re-founding* CDVC3 *on the full downset) buys anything over the closure-indexed form.*

Open Question 15.4 (the feasible update layer — resolved: from refuted conditionings to a conditioned metatheory)**.** *Condition the update-layer VCs on an update-and-merge-closed invariant and host the tombstone-free RGA. The required invariant machinery (*RgaInv $\wedge$ id_mono*, closed under merge given id-monotone allocation) is already mechanized on the data-type side — but the conditioning of convergence itself is subtler than "restrict the VCs to* Inv*-states," and the naive form is refuted (*`G2_Transport_Probe`*). Replacing* $\not\rightleftarrows$ *by its* Inv*-and-applicability-conditioned form inside* lo^E *drops order edges between events that are never jointly applicable (an* Ins *and a* Del *of the same node vacuously "commute"), and two orders that fold to different states then both respect* lo^E*: convergence fails. Strengthening* Inv *cannot repair it — conditioning is antitone, so a smaller applicability domain only removes more edges. The order must instead be applicability-aware: an edge survives when one event's applicability depends on the other. Restricting convergence to strictly applicability-valid enumerations is not the answer, though it looks strictly more general (*`G2_Applicability_Aware: separating_inequivalence`*): it is unsatisfiable for genuinely reachable versions carrying redundant concurrent ops — two concurrent deletes of one node, for which no enumeration keeps both deletes applicable. The resolution, and the finding, is that the two repairs are complementary, not competing: the correct notion is the applicability-aware order together with no-op-feasible enumerations (each step applicable or acting as identity). The order excludes the dependency-violating replays (which mere no-op tolerance would re-admit, breaking convergence); no-op tolerance admits the harmless redundant ones (which strict applicability rejects, breaking satisfiability). On the two decisive cases — one dependency, one redundancy — this notion is provably both convergent and satisfiable, where each single repair fails one (*`UpdateFeasibility_Gate: loOnA_noopFeasible_verdict`*). The general convergence theorem then factors cleanly (*`ConditionedConvergence`*): a single order VC ("a vacuous* commutesOn *edge is an applicability dependency") re-adds exactly the dropped edges, reducing conditioned convergence to the unconditioned theorem (*`conditioned_convergence_on`*) — for any data type whose commutation is available unconditionally. The tombstone-free RGA is precisely the one that is not (its commutation is reachability-conditioned), so it forces the conditioned route. There the last obstruction resolves sharply (*`RGA_Rehoming_Gate`*): the interleaved-applicability side-condition is false for the RGA — a concurrent delete rehomes nodes and stales a concurrent insert's path, and the staled insert is neither applicable nor a no-op — yet the version still converges, because the insert re-anchors at exactly the node the delete rehomed to (the RGA's own* `insdel_comm`*). So the obstruction is not unreachability, and not non-convergence: it is that the peel bubble demands pointwise applicability at hybrid fold states, while a path-carrying* *data type supplies semantic commutation only at the pair's common causal base. That architecture now exists, and §10–§14 of this note develop it in full: it is not a repaired bubble — the swap frame itself is refuted at merge unions — but a conditioned metatheory whose target is the raw fold up to ≈. The ladder above is the sequence of kernel-checked refutations that found it; the resolution is the end-to-end theorem* `rga_tombstone_free_ra_linearizable3_eq`.

Open Question 15.5 (the automation layer)**.** *The published catalogue's seventeen induction-scheme VCs were an SMT-facing Skolemization of feasibility, aimed at the wrong target. Design the analogous per-data-type induction scheme whose induction implies* CDVC3 *and the feasible delta laws — restoring push-button discharge, now of VCs that actually imply RA-linearizability. The observed uniformity of the discharges (σ-characterization by sandwich invariant + absorber trichotomy) suggests the scheme exists.*

Open Question 15.6 (criss-cross merges and virtual LCAs)**.** *The transition system gates Merge on the existence of an LCA (§4.1); criss-cross configurations are reachable, and every version in them linearizes, but their heads cannot merge — the model is silent exactly where git reaches for its recursive strategy (merge the maximal common ancestors into a virtual base, then merge with that). The metatheory looks ready for the extension: the ternary Join Lemma is stated over backward-closed event sets, and $E_1 \cap E_2$ needs no witnessing store version. What remains is a virtual-LCA Merge rule plus one per-data-type question: do the eight VCs imply that the recursively computed base equals $\sigma(E_1 \cap E_2)$? For the counter this is inclusion–exclusion once more — exact iff the maximal common ancestors jointly cover the intersection, itself a candidate store invariant; for the OR-set, a pointwise Boolean check. A positive answer closes the one visible gap between the model and practice.*

16 Mechanization

Everything above is mechanized in Lean 4 (Mathlib; axioms: `propext`, `Classical.choice`, `Quot.sound`; no `sorryAx` anywhere) in the Sal repository, under `Sal/ConditionedMRDTs/`:

Artifact	Lean	File
signature, ternary lo	`MRDTSig`, `ConditionedMRDTSig`	`Framework/MRDTSig.lean`
store, DAG, transitions	`Configuration`, `Step3`	`Framework/ExecutionModel.lean`, `Metatheory/LCA_Lemma.lean`
Lemma 4.1	`lca_events_of_storeInv`	`Metatheory/LCA_Lemma.lean`
σ/lo^E layer (merge-independent)	`UpdateVCs`, `IsCanonicalState`	`Framework/Sigma_LoOn3.lean`
the eight VCs	CoreVCs3CD, FeasibleDeltaVCs3, CDVC3	`Framework/VC_Set.lean`
Theorem 8.1 + bridges	`ra_linearizable_of_core_feasible_cd3`	`Metatheory/Adequacy.lean`
Table 1	all instance theorems	`MRDT_Instances/` (one directory per RDT)
impossibility kill-tests	—	`Refutations/Impossibility.lean`
findings journal T0–T11	—	`Development/MRDT_METATHEORY_DRAFT.md`
Conditioned-metatheory investigation (§4.2, Open Questions 15.3 & feasible-update; resolution in §10–§14):		
defeater vs. 2-OP (§4.2)	`no_inter_lca_2op_rem_peel_of_defeater`	`Refutations/InterLca2op_Defeater_Arbiter.lean`
peel obstruction (OQ 15.3)	`reunification_peel_obstruction`, `no_proper_back_block`	`Refutations/Reunification_Peel_Obstruction.lean`, `Metatheory/JoinLemma3C.lean`
closure-indexed contract	`JoinLemma3C`, `killTest_no_common_U`	`Refutations/JoinLemma3F_Of_AlmostClosed.lean`
closure-indexed adequacy + instances	`ra_linearizable3_of_joinC`, `ConditionedContract`	`Metatheory/ConditionedContract.lean`
all 9 discharges via the contract	`*_adequate_viaContract` (9)	`Development/AllMRDTs_viaContract.lean`
update-layer feasibility notion	`loOnA_noopFeasible_verdict`	`Refutations/UpdateFeasibility_Gate.lean`
conditioned convergence (order-half)	`conditioned_convergence_on`	`Framework/ConditionedConvergence.lean`
RGA rehoming: oracle false, converges	`verdict_oracle_false_but_converges`	`Refutations/RGA_Rehoming_Gate.lean`
feasible-update gate	`G2_conditioned_convergence_refuted`, `separating_inequivalence`	`Refutations/G2_{Transport_Probe,Applicability_Aware}.lean`
The conditioned metatheory (tombstone-free RGA end-to-end, feasible-update question resolved):		
merge-union witness refutation	—	`Development/RGA_HEnum_Refutation.lean`
observational quotient $D \mapsto D_{\approx}$	`QSig`, `qapplicable`	`Metatheory/GenericEqQuotient*.lean`
canonical-witness layer, raw-≈ target	`IsRALinearizable3Eq`, `RA_linearizable_up_to_eq_H`	`Metatheory/GoodConfig3H.lean`
RGA canonical engine	`CanonInv`, `canon_fold`	`MRDT_Instances/RGA_TombstoneFree/RGA_Canon*.lean`
generation discipline from born accuracy	`genDisc2C_of_born`	`MRDT_Instances/RGA_TombstoneFree/RGA_GenDisc_Assembly.lean`
capstone + discharged leaves	`rga_RA_linearizable_skeleton3`	`MRDT_Instances/RGA_TombstoneFree/RGA_Skeleton3.lean`, `RGA_*_Discharge.lean`
honest residual	`HonestDelivery`, `rga_RA_linearizable_honest`	`MRDT_Instances/RGA_TombstoneFree/RGA_Honest_Residual.lean`
mainline entry point	`rga_tombstone_free_ra_linearizable3_eq`	`MRDT_Instances/RGA_TombstoneFree/RA_Lin.lean`
flat collapse of the framework	`eqJoinH_of_joinC`, `flat_ra_linearizable3_eq`	`Metatheory/FlatGeneric_Bridge.lean`
the flat instances, generically	`*_ra_linearizable3_eq` (11)	`MRDT_Instances/ (per-RDT)`

The update-side layer (lo^E, convergence, σ) is merge-independent — it is stated over the update signature $\langle \Sigma, \sigma_0, \mathsf{do}, \mathsf{rc} \rangle$ alone — and is shared with the binary (CRDT) specialization of the theory; in the repository it lives in `Sal/CRDTs/Metatheory/`, the directory that also holds the binary exactness results quoted in Open Question 15.1. The production data types of Table 1 are mirrored faithfully from `Sal/MRDTs/*` (encoding deviations documented per instance); their original 24-VC discharges are untouched.

The foundations live under `Framework/`, the metatheorems under `Metatheory/`, the machine-checked negative results under `Refutations/`, and the per-RDT conditioned instances under `MRDT_Instances/` (the tombstone-free RGA chain in its `RGA_TombstoneFree/` directory); findings journals and superseded routes remain under `Development/`, which nothing imports. Each cited theorem is kernel-checked with axioms in {`propext`, `Classical.choice`, `Quot.sound`} (no `sorryAx`). They import one linearization file carrying two unrelated `sorry`s that the cited theorems do not transitively depend on; `Development/CONDITIONED_METATHEORY_PLAN.md` is their roadmap.